

nD: 2, H: 2, H+: 2

H0H1H+0H+1

$$-\left(\frac{1}{4}\right)M_s^{-2}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}H_iD_{\alpha_1}^{\dot{\alpha}_1}H_j(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (1)$$

H0H1H+0H+1

$$-\left(\frac{1}{4}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}H_iD_{\alpha_1}^{\dot{\alpha}_1}H_j(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (2)$$

H0H0H+1H+1

$$-\left(\frac{1}{4}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}H_iH_jD_{\alpha_1}^{\dot{\alpha}_1}(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (3)$$

nD: 0, H: 2, H+: 2

H0H0H+0H+0

$$+\left(\frac{1}{2}\hat{\mu}_h^2\right)\varepsilon_{il}\varepsilon_{kj}H_iH_j(H^\dagger)_k(H^\dagger)_l \quad (4)$$

nD: 0, H: 3, H+: 3

H0H0H0H+0H+0H+0

$$-\left(\frac{1}{8}\lambda_h\right)M_s^{-2}\varepsilon_{il}\varepsilon_{jm}\varepsilon_{kn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (5)$$

H0H0H0H+0H+0H+0

$$-\left(\frac{1}{4}\lambda_h\right)M_s^{-2}\varepsilon_{il}\varepsilon_{jn}\varepsilon_{mk}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (6)$$

H0H0H0H+0H+0H+0

$$+\left(\frac{1}{8}\lambda_h\right)M_s^{-2}\varepsilon_{im}\varepsilon_{jn}\varepsilon_{lk}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (7)$$

H0H0H0H+0H+0H+0

$$-\left(\frac{1}{8}\lambda_h\right)M_s^{-2}\varepsilon_{im}\varepsilon_{kn}\varepsilon_{lj}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (8)$$

H0H0H0H+0H+0H+0

$$+\left(\frac{1}{8}\lambda_h\right)M_s^{-2}\varepsilon_{in}\varepsilon_{jm}\varepsilon_{lk}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (9)$$

H0H0H0H+0H+0H+0

$$-\left(\frac{1}{2}\lambda_h\right)M_s^{-2}\varepsilon_{in}\varepsilon_{lj}\varepsilon_{mk}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (10)$$

nD: 0, H: 1, H+: 2, Q+: 1, uC+: 1

H0H+0H+0Q+0uC+0

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_u)^{uv}\varepsilon_{ij}\varepsilon_{kl}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,u}(u_\mathbb{C}^\dagger)_c^{\dot{\alpha}_2,v} \quad (11)$$

H0H+0H+0Q+0uC+0

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_u)^{uv}\varepsilon_{ik}\varepsilon_{lj}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,u}(u_\mathbb{C}^\dagger)_c^{\dot{\alpha}_2,v} \quad (12)$$

nD: 0, L: 1, eC: 1, H: 1, H+: 2

L0eC0H0H+0H+0

$$-\left(\frac{1}{4}\right)M_s^{-2}(y_e^\dagger)^{vu}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_1\alpha_2}L_{\alpha_1,i}^u(e_\mathbb{C})_{\alpha_2}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (13)$$

L0eC0H0H+0H+0

$$-\left(\frac{1}{4}\right)M_s^{-2}(y_e^\dagger)^{vu}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_1\alpha_2}L_{\alpha_1,i}^u(e_\mathbb{C})_{\alpha_2}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (14)$$

nD: 0, Q: 1, dC: 1, H: 1, H+: 2

Q0dC0H0H+0H+0

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_d^\dagger)^{vu}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^u(d_\mathbb{C})_{\alpha_2,bc}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (15)$$

Q0dC0H0H+0H+0

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_d^\dagger)^{vu}\varepsilon_{il}\varepsilon_{kj}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^u(d_\mathbb{C})_{\alpha_2,bc}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (16)$$

nD: 0, H: 2, H+: 1, L+: 1, eC+: 1

H0H0H+0L+0eC+0

$$+\left(\frac{1}{2}\right)M_s^{-2}(y_e)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_1,u}(e_\mathbb{C}^\dagger)_c^{\dot{\alpha}_2,v} \quad (17)$$

nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1

H0H0H+0Q+0dC+0

$$+\left(\frac{1}{4}\right)M_s^{-2}(y_d)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,u}(d_\mathbb{C}^\dagger)_c^{\dot{\alpha}_2,v} \quad (18)$$

nD: 0, Q: 1, uC: 1, H: 2, H+: 1

Q0uC0H0H0H+0

$$+\left(\frac{1}{4}\right)M_s^{-2}(y_u^\dagger)^{vu}\varepsilon_{ji}\varepsilon_{kl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^u(u_\mathbb{C})_{\alpha_2,bc}^vH_jH_k(H^\dagger)_l \quad (19)$$